

High-precision evaluation of Wigner's d matrix by exact diagonalization

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The precise calculations of Wigner's d matrix are important in various research fields. Due to the presence of large numbers, direct calculations of the matrix using Wigner's formula suffer from a loss of precision. We present a simple method to avoid this problem by expanding the d matrix into a complex Fourier series and calculate the Fourier coefficients by exactly diagonalizing the angular momentum operator J_y in the eigenbasis of J_z . This method allows us to compute the d matrix and its various derivatives for spins up to a few thousand. The precision of the d matrix from our method is about 10^{-14} for spins up to 100.

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I. INTRODUCTION

The spin is a fundamental quantum object and an important candidate for various quantum technologies, such as magnetic resonance spectroscopy, quantum metrology, and quantum information processing. An essential requirement in these developments is the precise control of many spins or alternatively a large spin composed of the constituent spins. The simplest case of such control is the rotation around a fixed axis. Accurately describing this process requires high-precision calculations of Wigner's d matrix [1–4] that quantifies the rotation of angle θ around the y axis: $d_{m,n}^j(\theta) \equiv \langle j, m | \exp(-i\theta J_y) | j, n \rangle \in \mathbb{R}$, where $|j, m\rangle$ is an eigenstate of J_z with eigenvalue m , i.e., $J_z |j, m\rangle = m |j, m\rangle$. Hereinafter, $\hbar = 1$ and $i = \sqrt{-1}$.

High-precision calculations of the d matrix is of interest in quantum metrology [5–7]. For instance, let us consider an atomic Ramsey (or equivalently, Mach-Zehnder) interferometer fed with all spins down as the paradigmatic setup of interferometric phase estimation. These spins then undergo an unknown phase shift θ via the evolution $\exp(-i\theta J_y)$ inside the interferometer. Finally, by detecting the population imbalance at the output port of the interferometer, i.e., the J_z measurement with respect to the output state $\exp(-i\theta J_y) |j, -j\rangle$, one can record $(2j + 1)$ possible outcomes. The outcome m occurs with probability $P_m(\theta) = |\langle j, m | \exp(-i\theta J_y) | j, -j \rangle|^2 = [d_{m,-j}^j(\theta)]^2$ conditioned on the unknown parameter θ , thus θ can be inferred from appropriate data processing on the outcomes. This process, however, requires accurate evaluation of Wigner's d matrix. In addition, the ultimate sensitivity of this estimation is determined by the Fisher information [5–7]: $F(\theta) \equiv \sum_m [\partial P_m(\theta) / \partial \theta]^2 / P_m(\theta)$, which requires accurate evaluation of the first-order derivative of Wigner's d matrix.

In addition to quantum metrology, Wigner's d matrix is closely related to spherical harmonics and Legendre polynomials and is of interest in many other fields [8–14]. However, the calculation of the d matrix for large spins ($j \gg 1$) suffers from a serious loss of precision due to the presence of large

numbers that exceed the floating-point precision in Wigner's original formula [1–4]. To avoid this problem, the d matrix has been calculated by means of recurrence relations [2]. This method still encounters severe numerical instability in the case of high spin, although a few remedies have been proposed [15–24]. Recently, Gumerov and Duraiswami [24] have developed a new recursion relation for each subspace of spins, which greatly improves the stability. However, the achievable precision (i.e., the maximum absolute error) of their results remains unclear. Most recently, Tajima [25] proposed Fourier-series expansion of Wigner's d matrix and converted the accurate evaluation of the d matrix to that of the Fourier coefficients. Such a Fourier-series representation has been shown to be more useful in improving the numerical stability and precision. However, each Fourier coefficient is still a sum of many large numbers that exceed the floating-point precision, so it has to be evaluated with the assistance of a formula-manipulation software [25].

In this paper, we put forward a very simple method to resolve the above large-number problem in evaluating the Fourier coefficients of Wigner's d matrix [25]. The essential idea is to express these coefficients via the inner products $\langle j, m | j, \mu \rangle_y$ where the eigenstates of J_y , i.e., $\{|j, \mu\rangle_y\}$ constitute an orthonormalized and completed set. To evaluate such inner products, we write down J_y as a Hermitian matrix in the eigenbasis of J_z . Then we numerically diagonalize the J_y matrix to obtain the eigenstates and the inner products. Due to the normalization of $|j, \mu\rangle_y$, the norm of each Fourier coefficient is not larger than unity, thus we avoid the large-number problem in floating-point calculations. This method allows us to evaluate accurately the d matrix and its various derivatives for much larger spins up to a few thousands with an absolute error of $O(10^{-14})$ for the d matrix and $O(j^k 10^{-14})$ for its k th-order derivative.

II. FOURIER SERIES OF WIGNER'S d MATRIX

For large spin j , numerical calculation of Wigner's formula is subject to intolerable numerical errors because it is a sum of

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many large numbers with alternating signs [1–4],

$$d_{m,n}^j(\theta) = \sum_k w_k^{(j,m,n)} \left(\cos \frac{\theta}{2} \right)^{2j-2k+n-m} \left(\sin \frac{\theta}{2} \right)^{2k+m-n}, \quad (1)$$

where

$$w_k^{(j,m,n)} = \frac{(-1)^{k+m-n} \sqrt{(j+m)!(j-m)!(j+n)!(j-n)!}}{(j-m-k)!(j+n-k)!(k+m-n)!k!},$$

and $k \in [\max(0, n-m), \min(j-m, j+n)]$. Taking $d_{0,0}^j(\pi/2)$ as an example, the term $k = j/2$ has a very large magnitude $|w_{j/2}^{(j,0,0)}|/2^j \propto 2^j/j$, exceeding the floating-point precision when $j \gg 1$.

To avoid this problem, Tajima [25] has proposed to expand Wigner's d matrix into a complex Fourier series,

$$d_{m,n}^j(\theta) = \sum_{\mu=-j}^{+j} e^{-i\mu\theta} t_{\mu}^{(j,m,n)}. \quad (2)$$

This representation of the d matrix is very useful and free from the large-number errors since each Fourier coefficient is less than or equal to 1 (see below). However, an accurate evaluation of the Fourier coefficients $t_{\mu}^{(j,m,n)}$ by means of Eq. (1) remains nontrivial. This is because the large-number problem still exists in the coefficients,

$$\begin{aligned} t_{\mu}^{(j,m,n)} &= \frac{1}{2\pi} \int_0^{2\pi} d_{m,n}^j(\theta) [e^{-i\mu\theta}]^* d\theta \\ &= \sum_{k=\max(0, n-m)}^{\min(j-m, j+n)} w_k^{(j,m,n)} I_{\mu}(2j, 2k+m-n), \end{aligned} \quad (3)$$

where we have introduced an integral,

$$\begin{aligned} I_{\mu}(2j, \lambda) &\equiv \frac{1}{2\pi} \int_0^{2\pi} \left(\cos \frac{\theta}{2} \right)^{2j-\lambda} \left(\sin \frac{\theta}{2} \right)^{\lambda} e^{i\mu\theta} d\theta \\ &= \frac{1}{2^{2j}} \sum_{l=\max\{0, -j+\mu+\lambda\}}^{\min\{\lambda, j+\mu\}} (-1)^{l-\lambda/2} \binom{2j-\lambda}{j+\mu-l} \binom{\lambda}{l}, \end{aligned}$$

with $\binom{\lambda}{l} = \lambda!/[l!(\lambda-l)!]$. When $j \gg 1$, some terms in the integral are still huge (e.g., the terms $\lambda = j$, $l = 0$, and $\mu = -j/2$). Tajima [25] bypassed this problem by employing a symbolic computation software and then reducing the results to double-precision floating point numbers.

III. METHOD OF EXACT DIAGONALIZATION

Here, instead of using Eq. (3), we present a very simple method to calculate the Fourier coefficients $t_{\mu}^{(j,m,n)}$ that is free from the above-mentioned large-number problem. The key observation is that the d matrix can be rewritten as

$$\begin{aligned} d_{m,n}^j(\theta) &\equiv \langle j, m | e^{-i\theta J_y} | j, n \rangle \\ &= \sum_{\mu=-j}^{+j} e^{-i\mu\theta} \langle j, m | j, \mu \rangle_{yy} \langle j, \mu | j, n \rangle, \end{aligned} \quad (4)$$

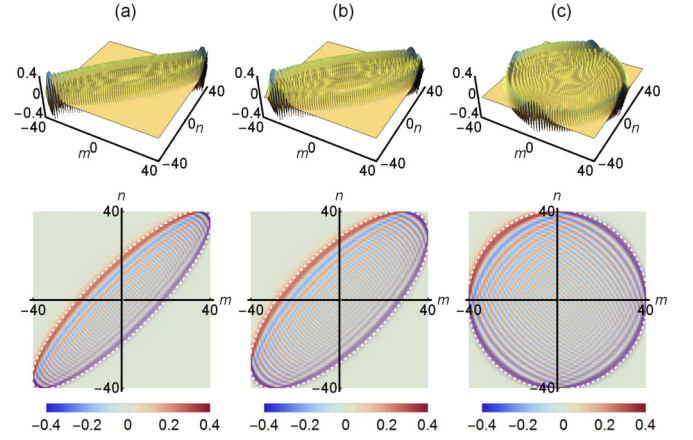


FIG. 1. (Color online) Computed results of Wigner's d matrix $d_{m,n}^j(\theta)$ (with total spin $j = 40$) against m and n for (a) $\theta = \pi/6$, (b) $\pi/4$, and (c) $\pi/2$, respectively. The dashed lines in bottom panel are the boundary of the central region, determined by Eq. (9). Outside the region, the values of $d_{m,n}^j(\theta)$ are almost vanishing.

where $|j, \mu\rangle_y \equiv e^{i(\pi/2)J_x} |j, \mu\rangle = e^{-i(\pi/2)J_z} e^{-i(\pi/2)J_y} e^{i(\pi/2)J_z} |j, \mu\rangle$ are eigenstates of J_y and they constitute an orthonormalized and completed set, i.e., ${}_y\langle j, \mu | j, \mu' \rangle_y = \delta_{\mu, \mu'}$ and $\sum_{\mu} |j, \mu\rangle_y {}_y\langle j, \mu| = 1$. Hereafter, we use $|j, m\rangle$ for the eigenstates of J_z and $|j, \mu\rangle_y$ for the eigenstates of J_y . Comparing Eqs. (2) and (4), we identify the Fourier coefficients in Eq. (2) as

$$\begin{aligned} t_{\mu}^{(j,m,n)} &= \langle j, m | j, \mu \rangle_{yy} \langle j, \mu | j, n \rangle \\ &= e^{i(\pi/2)(n-m)} d_{m,\mu}^j \left(\frac{\pi}{2} \right) d_{n,\mu}^j \left(\frac{\pi}{2} \right), \end{aligned} \quad (5)$$

which obeys the sum rule $\sum_{\mu} t_{\mu}^{(j,m,n)} = \langle j, m | j, n \rangle = \delta_{m,n}$. From Eq. (5), one can note that all the Fourier coefficients and hence the d matrix for arbitrary θ depend on $d_{m,\mu}^j(\theta = \pi/2)$ only [17,26,27]. The elements of the d matrix are real and obey the following symmetry properties (see Refs. [1,25], and Fig. 1):

$$\begin{aligned} d_{m,n}^j(\theta) &= d_{m,n}^j(-\theta) = d_{-m,-n}^j(\theta) = (-1)^{n-m} d_{m,n}^j(\theta), \\ d_{m,n}^j(\pi - \theta) &= (-1)^{j+m} d_{m,-n}^j(\theta) = (-1)^{j-n} d_{-m,n}^j(\theta), \\ d_{m,n}^j(\pi + \theta) &= (-1)^{j-n} d_{m,-n}^j(\theta) = (-1)^{j+m} d_{-m,n}^j(\theta), \end{aligned}$$

from which one can easily obtain $t_{-\mu}^{(j,m,n)} = (-1)^{2j+m+n} t_{\mu}^{(j,m,n)}$ and $t_{\mu}^{(j,n,m)} = t_{\mu}^{(j,-m,-n)} = (-1)^{n-m} t_{\mu}^{(j,m,n)}$ as observed recently by Tajima [25].

Most importantly, the first result of Eq. (5) provides a very simple but accurate method to calculate the Fourier coefficients and hence Wigner's d matrix by solving the inner product $\langle j, m | j, \mu \rangle_y$. To this end, we first express J_y as a $(2j+1) \times$

dimensional Hermitian matrix,

$$J_y = \frac{1}{2i} \begin{pmatrix} 0 & -X_{-j+1} & & & \\ X_j & 0 & -X_{-j+2} & & \\ & X_{j-1} & 0 & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & 0 & -X_j \\ & & & X_{-j+1} & 0 \end{pmatrix}, \quad (6)$$

where the matrix elements are determined by $\langle j, m | J_y | j, n \rangle = (X_{-n} \delta_{m, n+1} - X_n \delta_{m, n-1}) / (2i)$ with the term $X_m = \sqrt{(j+m)(j-m+1)}$ satisfying $X_{\pm m} = X_{\mp m+1}$ and $X_{-j} = 0$. Next, we diagonalize the Hermitian matrix using standard numerical methods, e.g., the ZHBEV subroutine of LAPACK, or the EVCHF package of the IMSL, to obtain all the eigenvectors $\{|j, \mu\rangle_y\}$ and their probability amplitudes $\langle j, m | j, \mu \rangle_y$. For the simplest case $j = 1/2$, the matrix is indeed the transpose of the Pauli matrix σ_y over 2 (i.e., $\sigma_y^T/2$), which gives two eigenvectors $|1/2, \pm 1/2\rangle_y$, corresponding to the eigenvalues $\pm 1/2$. The inner products $\langle j, m | j, \mu \rangle_y$ for $j = 1/2$ are given by $\langle 1/2, 1/2 | 1/2, \pm 1/2 \rangle_y = 1/\sqrt{2}$ and $\langle 1/2, -1/2 | 1/2, \pm 1/2 \rangle_y = \pm i/\sqrt{2}$. Inserting them into Eq. (4), one can obtain Wigner's d matrix.

The exact-diagonalization method has two advantages. First, the magnitude of $\langle j, m | j, \mu \rangle_y$ and hence all the coefficients $t_\mu^{(j, m, n)}$ in Eq. (5) are not larger than unity since all the eigenvectors $\{|j, \mu\rangle_y\}$ are normalized. This provides a solution to the large-number problem in Eqs. (1) and (3). Second, the matrix in Eq. (6) is tridiagonal and Hermitian, which can be easily diagonalized. By diagonalizing the J_y matrix only once, all the Fourier coefficients and hence all the elements of the d matrix can be obtained with given j and θ (see the Supplemental Material [28]). This method allows us to calculate Wigner's d matrix for j up to a few thousand. More importantly, one can compute the arbitrary k th-order derivative of the d matrix with little additional cost due to

$$\begin{aligned} \frac{\partial^k d_{m,n}^j(\theta)}{\partial \theta^k} &= \langle j, m | e^{-i\theta J_y} (-i J_y)^k | j, n \rangle \\ &= \sum_{\mu=-j}^{+j} (-i\mu)^k e^{-i\mu\theta} t_\mu^{(j, m, n)}. \end{aligned} \quad (7)$$

This is indeed a common advantage of the Fourier-series representation. As a comparison, one can find that the direct evaluation of the first result of Eq. (7) costs double even for the case of $k = 1$ since $\langle j, m | \exp(-i\theta J_y) (-i J_y) | j, n \rangle$ gives

$$\frac{\partial d_{m,n}^j(\theta)}{\partial \theta} = \frac{1}{2} [X_n d_{m,n-1}^j(\theta) - X_{-n} d_{m,n+1}^j(\theta)], \quad (8)$$

which depends on two elements of the d matrix. When $n = \pm j$, only one of them remains.

IV. NUMERICAL RESULTS AND ERROR ANALYSIS

As shown in Fig. 1, we plot the computed results of $d_{m,n}^j(\theta)$ on the plane (m, n) with $m, n \in [-j, +j]$. For a relatively large spin $j = 40$ and a given θ , $d_{m,n}^j(\theta)$ is appreciable only in the

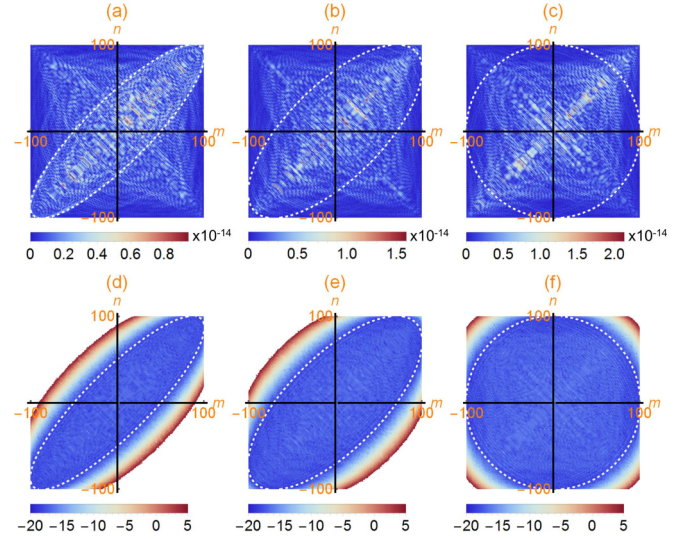


FIG. 2. (Color online) The absolute error $\Delta_{\text{abs}} = |(d_{\text{comp}} - d_{\text{ex}})|$ (the upper panel) and the relative error $\log_{10} \Delta_{\text{rel}}$ (the lower panel) of Wigner's d -matrix $d_{m,n}^j(\theta)$ with the spin $j = 100$ and the rotation angle $\theta = \pi/6$ (left), $\pi/4$ (middle), and $\pi/2$ (right), respectively. The computed results are obtained by diagonalizing the matrix in Eq. (6), using the ZHBEV subroutine of LAPACK; whereas the exact results d_{ex} are obtained by *Mathematica 10.0*. The dashed lines are given by Eq. (9).

central region and tends to zero quickly outside this region. The boundary of this region is determined by (see also the dashed lines of Fig. 2)

$$m^2 + n^2 - 2mn \cos \theta = j(j+1) \sin^2 \theta, \quad (9)$$

at which $\partial^k d_{m,n}^j(\theta) / \partial \theta^k = 0$ for $k = 1, 2$. This boundary equation follows from the differential equation of the d matrix [2]. A similar boundary equation has been obtained using the WKB approximation [29].

Given the exact value d_{ex} and the numerically calculated value d_{comp} of a matrix element $d_{m,n}^j(\theta)$, the absolute error is defined as $\Delta_{\text{abs}} = |d_{\text{comp}} - d_{\text{ex}}|$, and the relative error is defined as $\Delta_{\text{rel}} = |(d_{\text{comp}} - d_{\text{ex}}) / d_{\text{ex}}|$. The exact values of the d -matrix elements are obtained by *Mathematica 10.0*. Figure 2 shows that the absolute error is $\Delta_{\text{abs}} \sim 10^{-14}$ even for a relatively large spin $j = 100$ (see also Fig. 3). The lower panel of Fig. 2 shows the relative error $\lesssim 10^{-10}$ within the central region (enclosed by the dashed line) but increases rapidly outside this region.

Outside the boundary, the large relative error simply follows from the very small exact values $|d_{\text{ex}}|$. To illustrate this point, we consider $d_{m,n}^j(\theta)$ with $|m| = |n| = +j$. In this case, we have an analytical expression,

$$d_{j,m}^j(\theta) = (-1)^{j-m} \binom{2j}{j+m}^{1/2} \left(\cos \frac{\theta}{2} \right)^{j+m} \left(\sin \frac{\theta}{2} \right)^{j-m}.$$

Using the symmetry, one can obtain $d_{\pm j, \mp j}^j(\theta) = [-\sin(\theta/2)]^{2j}$. For $j = 100$ and $\theta = \pi/6$, one can easily obtain the exact values $d_{\pm j, \mp j}^j(\pi/6) = 3.974 \times 10^{-118}$, which lie outside the boundary [see the dashed line of Fig. 2(d)]. By contrast, although the numerically calculated values

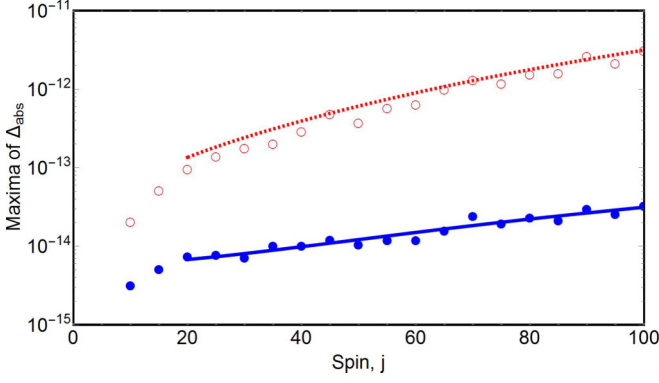


FIG. 3. (Color online) The maximum absolute error of $d_{m,m}^j(\theta)$ (solid circles) and that of $\partial d_{m,m}^j(\theta)/\partial\theta$ (open circles) as functions of spin j . Blue solid line: the fitting result, $\Delta_{\text{abs,max}} \approx (aj^2 + b) \times 10^{-14}$ with $a = 2.568 \times 10^{-4}$ and $b = 0.5758$. The precision of the d matrix can reach 3.275×10^{-14} at $j = 100$. Red dashed line: maximum error of the first-order derivative $\sim j \times \Delta_{\text{abs,max}}$.

$d_{\pm j, \mp j}^j(\pi/6) \sim 10^{-17}$ are very close to zero, they are much larger than the exact values, leading to a large relative error Δ_{rel} .

Due to the same reason, the d -matrix elements for other values of θ also show large Δ_{rel} outside the central region. For example, $\theta = \pi/2$, the boundary of the central region is a circle: $m^2 + n^2 = j(j+1)$ with a radius of $\sim j$. The d -matrix elements with $|m| = |n| = j$ have the same exact value $d_{m,n}^j(\pi/2) = 1/2^j \sim 7.9 \times 10^{-31}$, which is much smaller than the numerically calculated value of $\sim 10^{-15}$, yielding a large Δ_{rel} . The computed $d_{m,n}^j(\pi/2)$ at $mn = 0$ also shows large relative error. To see it clearly, we use the exact result [1],

$$d_{m,0}^j(\theta) = (-1)^m \sqrt{\frac{(j-m)!}{(j+m)!}} P_j^m(\cos \theta),$$

where $P_j^m(x)$ is the associated Legendre polynomial. When $\theta = \pi/2$, we obtain the exact results $d_{m,0}^j(\pi/2) = (-1)^m d_{0,m}^j(\pi/2) \propto P_j^m(0) = 0$ for odd $j - m$. In contrast, the computed results are small but nonzero, thus $\Delta_{\text{rel}} \rightarrow \infty$.

Finally, we discuss the scaling of the error in evaluating the d matrix and its various derivatives with increasing spin j . For this purpose, we sweep m , n , and θ and calculate the maximum absolute errors $\Delta_{\text{abs,max}}$ for $d_{m,n}^j(\theta)$ and $\partial d_{m,n}^j(\theta)/\partial\theta$ as functions of j for j up to 100. Since the maximum absolute

error of the d matrix almost appears inside the boundary (see the upper panel of Fig. 2), we only sweep (m,n) within the central region and increase θ from 0 to $\pi/2$ with an increment of $\pi/36$ [25]. We find that typically the maximum absolute error appears at $m \sim n$ and $\theta \sim 0$ or $\pi/2$. As shown in Fig. 3, one can find that the numerical results of $10^{14} \times \Delta_{\text{abs,max}}$ can be well fitted by $aj^2 + b$ with $a \sim 10^{-4}$ and $b \sim 0.6$ (see the solid line). This precision is slightly worse than the previous one $\Delta_{\text{abs,max}} \approx 10^{-14.8+0.006j}$ for spin j up to 100 [25]. However, if the scaling persists to larger j , our method could provide a smaller error as $j > 405$. One can also note that the maximum absolute error of $\partial d_{m,n}^j(\theta)/\partial\theta$ can be approximated by $j \times \Delta_{\text{abs,max}}$ (the dashed line). More generally, from Eqs. (4) and (7), we can deduce that the maximum absolute error in evaluating the k th-order derivative $\partial^k d_{m,n}^j(\theta)/\partial\theta^k$ is larger than that of the d matrix by a factor of $O(j^k)$.

V. CONCLUSION

To summarize, we have presented a very simple method to evaluate accurately Wigner's d matrix by diagonalizing the angular momentum operator J_y in the eigenbasis of J_z . The coefficients of the Fourier-series expansion of the d matrix, closely related to the eigenstates of J_y , are shown to be not larger than unity. This enables us to avoid the large-number problem. The absolute error of $d_{m,n}^j(\theta)$ can reach $\sim 10^{-14}$ for spin j up to 100 and the relative error of $\sim 10^{-10}$ within the central region $m^2 + n^2 - 2mn \cos \theta \leq j(j+1) \sin^2 \theta$, outside which the values of the d matrix tend to zero quickly. As one of the main advantages, we show that for given j and θ , all the elements of Wigner's d matrix and their k th-order derivatives can be obtained by diagonalizing J_y only once [28]. The method presented here is a part of the larger class of algorithms based on the matrix exponents [30]. As the matrix exponent of J_y , Wigner's d matrix could be computed even more accurately and efficiently since the matrix in Eq. (6) is Hermitian and tridiagonal.

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